

# Topic 11 – Alternative Splice Variants

## Exons, Introns, UTRs



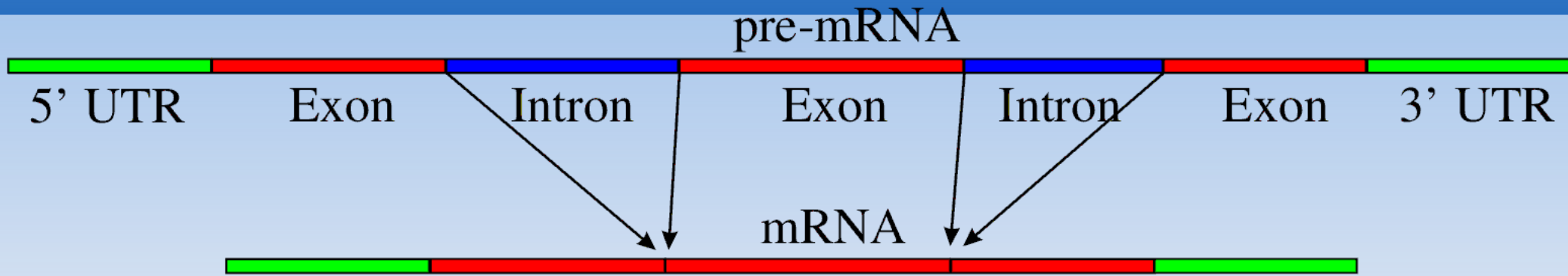
# Compare with an eukaryotic gene

Find the NIH Gene database entry for **human KCNH2 gene**

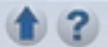
The screenshot shows the Entrez Gene database interface. At the top, the 'Entrez Gene' logo is visible. Below it, a search bar contains the text 'for KCNH2'. To the right of the search bar are buttons for 'Go', 'Clear', and 'Save Search'. Below the search bar, there are tabs for 'Limits', 'Preview/Index', 'History', 'Clipboard', and 'Details'. The 'Display' section shows 'Summary' selected, with a 'Show' button and a 'Send to' button. Below this, there are buttons for 'All: 24', 'Current Only: 24', 'Genes Genomes: 23', and 'SNP GeneView: 13'. The 'Items 1 - 20 of 24' section shows 'Page 1 of 2'. The first item is '1: [KCNH2](#)'. The details for KCNH2 are listed below:

**Official Symbol** KCNH2 and **Name:** potassium voltage-gated channel, subfamily H (eag-related), member 2 [*Homo sapiens*]  
**Other Aliases:** ERG1, HERG, HERG1, Kv11.1, LQT2, SQT1  
**Other Designations:** cause of Long QT Syndrome Type 2; ether-a-go-go-related potassium channel protein; human eag-related; potassium channel HERG; potassium channel HERG1; potassium voltage-gated channel, subfamily H, member 2; voltage-gated potassium channel; voltage-gated potassium channel, subfamily H, member 2  
**Chromosome:** 7; **Location:** 7q35-q36  
**Annotation:** Chromosome 7, NC\_000007.12 (150272982..150305947, complement)  
**MIM:** 152427

# Alternative Splicing

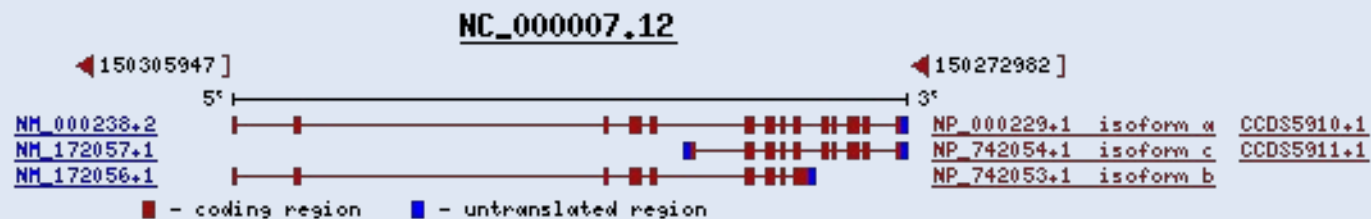


## Genomic regions, transcripts, and products



(minus strand) Go to [reference sequence details](#)

[Try our new Sequence Viewer](#)



## Genomic context



chromosome: 7; Location: 7q35-q36

[See KCNH2 in MapViewer](#)



# Alternative Splicing

- Humans have about 20,000 genes
- And about 100,000 distinct proteins!
  - on average, each gene may be capable of giving us 5 splice variants!
- We have tissue/organ-specific splicing factors that give us different mature mRNAs in different organs or tissues

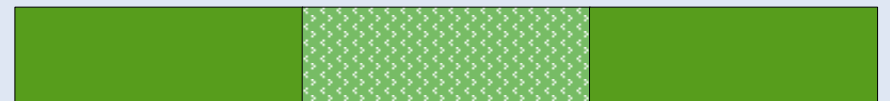
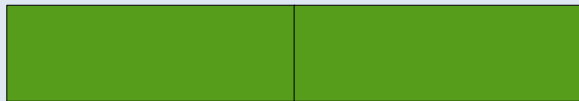
# Splice Variant mRNAs

- Suppose we have 2 splice variants of a gene

Pre-mRNA



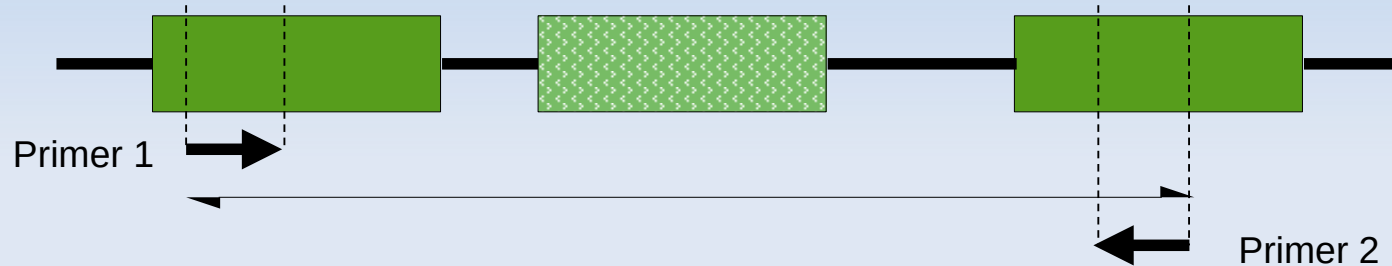
Mature  
mRNAs



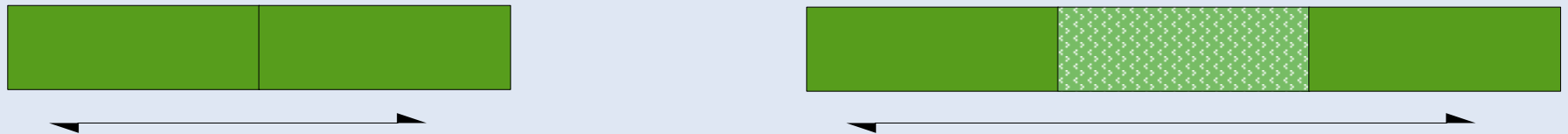
# Detecting Splice Variant mRNA

- We can design primers to detect splice variant expression

Pre-mRNA

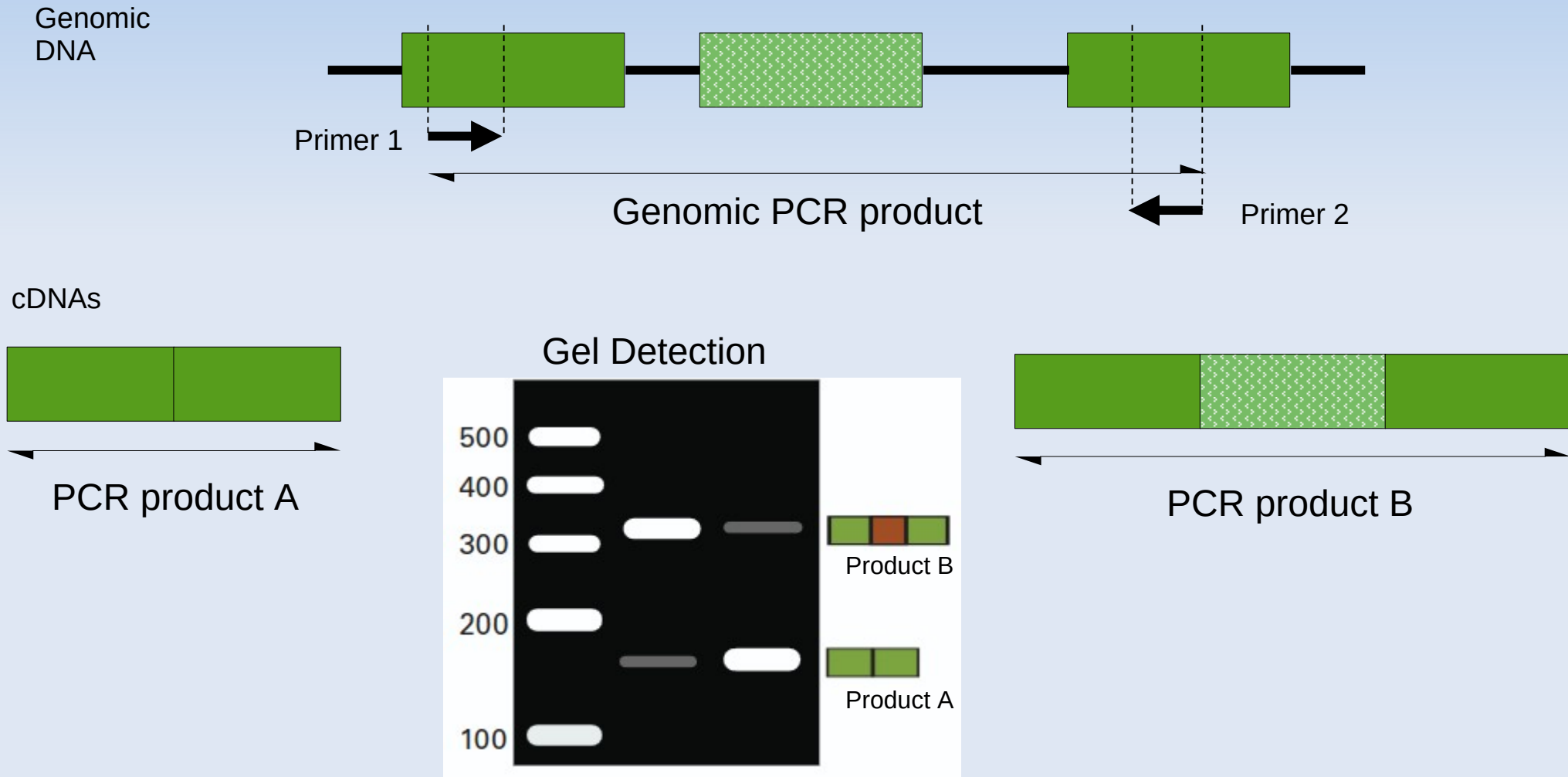


Mature mRNA



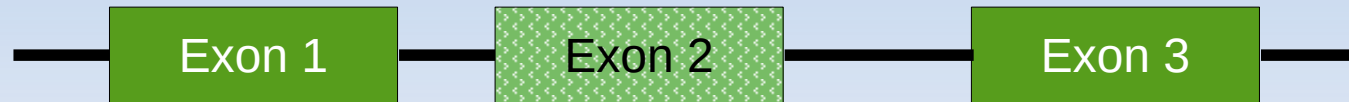
# Detecting Splice Variant mRNA

- PCR can give us 3 products:

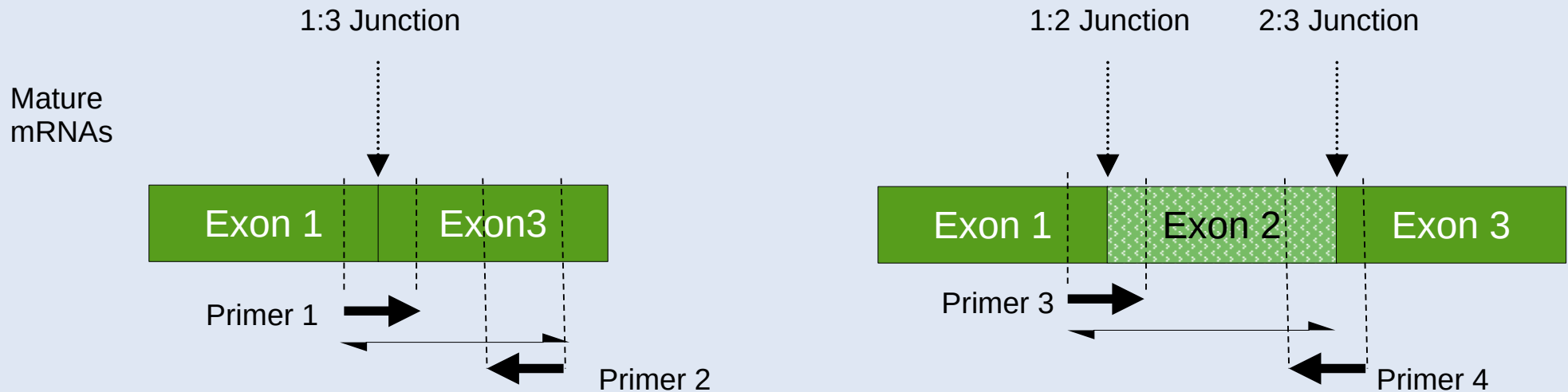


# Exon-exon Junction Primers

- We can target primers to the exon-exon junctions



Pre-mRNA



Mature mRNAs



# Exon-intron Junction Primers

- Primers for exon-intron junctions

